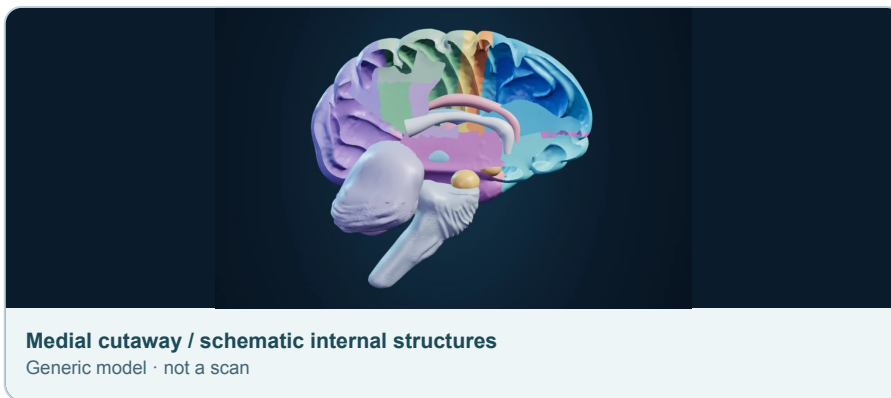


Scientific Report

An auditable explanation of the research guide and your reflection—not a scientific assessment of your individual brain.



26

Teaching regions

15

Core region studies

9/16

Present-day responses

OBSERVATION

What you directly reported. Not independently verified or objectively tested.

RESEARCH

What a cited study measured in its participants, with design and limits preserved.

HYPOTHESIS

Editorial combinations to consider. No calibrated confidence or established neural mechanism.

UNKNOWN

Your brain anatomy, causation, unreported traits, clinical status and actual synergy.

Correlation is not causation. A group association does not establish an individual effect. More or less volume, thickness, activity or connectivity is not a general scale of better or worse.

01 TRANSPARENT ARCHITECTURE

How this report was assembled

1. **Preserve observations.** History, reported developmental stage, response and support at the time are separate observations. Optional calendar ages never determine maturation. Prenatal context remains separate.
2. **Select research reading.** Endorsed history questions select existing question-to-study links. A domain match is different from related evidence; unmapped answers stay unmapped.
3. **Describe present-day patterns.** Only “Often” or “Sometimes” on a separate current observation selects a strength, friction or resource card. This is an editorial inclusion rule, not a validated cutoff.
4. **Suggest combinations.** Both listed strength observations must be endorsed. The output is an untested hypothesis. No study-derived interaction coefficient, risk estimate or neural parameter is computed.
5. **Render the same model.** Every reader sees the same generic teaching anatomy. Colors and separated pieces identify structures, not individual injury or activation.

Extrapolation stops before anatomy

“You reported noticing details” → “Signal scout is a possible shorthand” is a transparent editorial description. Adding “you also reported generating ideas” permits a possible combination to explore. It does **not** permit “therefore your amygdala is enlarged,” “your brain has a deformation,” or “you have a superior ability.” Adding the word “may” would not supply the missing evidence.

What could explain a similar answer?

Practice, personal interests, current circumstances, sleep, task demands, social support, reporting differences and many other influences can produce similar observations. The tool does not distinguish those explanations or establish that childhood events caused a present-day pattern.

What would stronger evidence require?

A prespecified construct and outcome, suitable validated measurements, an appropriate comparison, assessment of alternative explanations, independent replication and tests of generalization. A claimed neural mechanism would need relevant brain data; a casual questionnaire cannot substitute for that.

Methods context: Marek et al. (2022), *Reproducible brain-wide association studies require thousands of individuals*. DOI: [10.1038/s41586-022-04492-9](https://doi.org/10.1038/s41586-022-04492-9). Brain-wide association reproducibility is design-dependent; this does not mean all imaging studies need the same sample size. Publisher correction: 10.1038/s41586-022-04692-3.

Product validation status: The reflection and named combinations are not clinically validated instruments. The research library is curated, not a systematic review of all literature. The teaching model has not received an independent neuroanatomical expert review.

MATURATION-LED RESEARCH GUIDE

Sensitive periods: developmental processes, not age bands

A sensitive period concerns changing responsiveness in a particular system, to a particular experience or stimulus. This guide is organized by developmental process. It does not assign an individual opening or closing date.

Reported context

Developmental stage, response, support, and source.

Research measurement

Puberty, hormones, circuit properties, or task response.

Individual inference

Not established without suitable measurements and validation.

Pubertal development is informative, but is not identical to brain maturation. A birthday, gender identity, skill, diagnosis, or recalled milestone cannot substitute for a region-specific brain measurement. Calendar ages remain optional historical notes and never select a sensitive window in this app.

Threat, regulation & stress-system plasticity

Caregiving context and pubertal transition may matter for particular stress-response circuits. A reported stage can guide reading, but does not locate your circuit on a maturation curve.

A candidate sensitive window for a specified system and environment—not a universal “trauma window.”

[Gee et al. \(2013\)](#) [Gunnar et al. \(2019\)](#)

Affective reactivity & regulatory recruitment

Pubertal development, the type of stimulus and the regulation task are separate variables. Sex-related contrasts can change direction across conditions.

Task response does not establish a structural deformation, a fixed personality or a demonstrated power.

[Vijayakumar et al. \(2019\)](#) [Church et al. \(2025\)](#)

Regional structural maturation

Hippocampal, amygdala, cortical and striatal development need not follow the same trajectory. Age and pubertal measures can each contribute information in research models.

Physical puberty is a developmental proxy, not a measurement of every brain region or proof of a closed learning window.

[Goddings et al. \(2014\)](#) [Satterthwaite et al. \(2014\)](#)
[Wang et al. \(2026; online March\)](#)

Sex, gender & social experience

Assigned sex, hormone measurements, gender identity, gendered experience and social support are not interchangeable. Reported gender-related experiences remain part of the context.

No identity is mapped to a neural type, expected strength or presumed impairment.

[Dhamala et al. \(2024\)](#) [Church et al. \(2025\)](#)

Read the specific stimulus–response relationship

For example, Church et al. compared negative and neutral images under different regulation instructions. Some sex-related contrasts changed direction between conditions. That supports recording the actual task and stimulus, not assigning a general “more sensitive” sex.

Church et al. (2025): primary study

A significant association in one group and a nonsignificant association in another does not by itself demonstrate a group difference. Sex-stratified morphology is not a stimulus-response experiment. Gender, assigned sex and hormone measures are kept distinct. The studies retain their actual age covariates and limitations even though the questionnaire is stage-first.

Longitudinal structural MRI

Goddings et al. (2014)

DOI: 10.1016/j.neuroimage.2013.09.073 · Review: 2026-09-12

PARTICIPANTS

275 participants; 711 scans; study ages 7–20.

DEVELOPMENTAL MEASUREMENT

Self-reported Tanner stage plus chronological age, modeled separately and together.

STIMULUS / EXPOSURE

No stimulus task; structural development.

OUTCOME ACTUALLY MEASURED

Subcortical volumes, including amygdala, hippocampus and striatal regions.

INTERACTION OR STRATIFICATION

Age and puberty had independent and interacting associations; models were fitted separately by sex.

Finding: Different regions followed different developmental trajectories. Puberty did not replace every contribution of age.

Sex / gender interpretation: Separate female/male models; not a universal between-sex effect.

Limits: Observational, physical-stage proxy; not a whole-brain maturity meter or an individual sensitive-window test.

Cross-sectional structural MRI

Satterthwaite et al. (2014)

DOI: 10.1016/j.jaac.2013.12.002 · Review: 2026-09-12

PARTICIPANTS

524 pre- or postpubertal participants; study ages 10–22.

STIMULUS / EXPOSURE

No stimulus task; anatomy comparison.

INTERACTION OR STRATIFICATION

A puberty-by-sex interaction was detected in hippocampus, but not amygdala.

Finding: Postpubertal females had larger scaled hippocampal volumes than postpubertal males; prepubertal groups did not differ significantly.

Sex / gender interpretation: Tested puberty-by-sex interaction, region-specific.

Limits: Not a longitudinal change estimate. Volume is not ability; findings cannot be generalized to every region. Prior studies differed.

DEVELOPMENTAL MEASUREMENT

Pre/postpubertal classification; age and age-by-sex modeled as covariates.

OUTCOME ACTUALLY MEASURED

Hippocampal and amygdala volume/shape, scaled for intracranial volume.

Three-wave longitudinal task fMRI

Vijayakumar et al. (2019)

DOI: 10.1016/j.cortex.2019.04.024 · Review: 2026-09-12

PARTICIPANTS

82 adolescents; study ages 9–18.

STIMULUS / EXPOSURE

Passive viewing of emotional facial expressions.

INTERACTION OR STRATIFICATION

Sex-related trajectories varied by region. Some hormone analyses were within-sex rather than direct between-sex tests.

Finding: Exploratory models suggested mid-pubertal peaks in amygdala/hippocampal responses to fearful faces; testosterone did not explain all puberty-related changes.

Sex / gender interpretation: Exploratory, task-specific findings; within-group significance is not proof of a group difference.

Limits: Some condition-specific results did not survive multiple-comparison correction. Replication needed; not a validated personal sensitivity window.

DEVELOPMENTAL MEASUREMENT

Pubertal Development Scale, salivary testosterone and age considered separately.

OUTCOME ACTUALLY MEASURED

Task-related brain activation; not tissue size or injury.

Three-session longitudinal stress study

Gunnar et al. (2019)

DOI: 10.1073/pnas.1909699116 · Review: 2026-09-12

PARTICIPANTS

129 previously institutionalized and 170 comparison participants.

STIMULUS / EXPOSURE

Trier social stress test for children.

INTERACTION OR STRATIFICATION

Group-by-within-person pubertal-stage interaction, controlling for sex.

Finding: Pubertal progression accompanied increasing cortisol reactivity in previously institutionalized youth in supportive adoptive homes, not comparison youth.

Sex / gender interpretation: Sex was a covariate, not evidence for a sex-specific recalibration rule.

Limits: Supports a possible recalibration window in this context; does not demonstrate repaired tissue, a universal support offset or a sex-specific treatment.

[Published author correction](#)

DEVELOPMENTAL MEASUREMENT

Nurse-assessed pubertal stage at each annual session.

OUTCOME ACTUALLY MEASURED

Cortisol stress reactivity; not brain structure.

Cross-sectional emotion-regulation fMRI

Church et al. (2025)

DOI: 10.1016/j.dcn.2025.101588 · Review: 2026-09-12

PARTICIPANTS

180 early adolescents; 98 female and 82 male by parent-reported sex assigned at birth.

STIMULUS / EXPOSURE

Negative or neutral images, with instructions to react naturally or reappraise.

INTERACTION OR STRATIFICATION

Tested sex × stimulus valence × regulation interaction.

Finding: Female participants showed greater OFC recruitment during regulation of negative images; a bilateral/left OFC cluster showed the reverse contrast for neutral images.

Sex / gender interpretation: Measured assigned sex, not gender identity. Direction depends on task and stimulus.

Limits: Cross-sectional, task-specific. Amygdala follow-ups did not establish a simple sex difference within negative or neutral conditions. No individual vulnerability or ability estimate follows.

DEVELOPMENTAL MEASUREMENT

Recruitment sought similar pubertal status; findings checked with and without age adjustment.

OUTCOME ACTUALLY MEASURED

Task activation in orbitofrontal/middle-frontal regions and amygdala.

ABCD functional-connectivity modeling

Dhamala et al. (2024)

DOI: 10.1126/sciadv.adn4202 · Review: 2026-09-12

PARTICIPANTS

4,757 children, study ages 9–10; binary assigned-sex categories in the analyzed sample.

DEVELOPMENTAL MEASUREMENT

No individual maturity or sensitive-period measure.

STIMULUS / EXPOSURE

Resting-state connectivity; not a stimulus-response experiment.

OUTCOME ACTUALLY MEASURED

Associations with assigned sex and separate child/parent gender questionnaires.

INTERACTION OR STRATIFICATION

Sex and gender were analyzed as distinct constructs.

Finding: Within-sex models explained about 0.55–0.56% of parent-reported gender variance; self-reported gender prediction was not significant.

Sex / gender interpretation: Assigned sex and gender are distinct; broad gender generalizations are unsupported.

Limits: Gender measures were not equivalent to identity labels. Results do not establish gender-specific responses, classify someone’s identity, or validate a male/female brain type.

ENIGMA structural MRI mega-analysis

Wang et al. (2026; online March)

DOI: 10.1016/j.biopsych.2026.02.016 · Review: 2026-09-12

PARTICIPANTS

3,711 participants across 25 sites; clinical and comparison groups.

DEVELOPMENTAL MEASUREMENT

Age at scanning and recorded sex strata; not measured pubertal stage at exposure.

STIMULUS / EXPOSURE

Retrospective abuse/neglect exposure, not an experimental stimulus.

OUTCOME ACTUALLY MEASURED

Volume, cortical thickness and surface-area deviations.

INTERACTION OR STRATIFICATION

Age- and sex-stratified associations; not a universal sex multiplier.

Finding: Associations differed across sex, region and measurement. Stronger patterns were reported in young female adults; pediatric findings were not significant.

Sex / gender interpretation: Recorded female/male strata; individual comparability remains unknown.

Limits: Scan age is not exposure stage. Structural findings cannot be relabeled stimulus responses or personal sensitive periods. Abstract review only; sex ascertainment is not assumed.

Previously institutionalized youth comparison

Gee et al. (2013)

DOI: 10.1073/pnas.1307893110 · Review: 2026-09-12

PARTICIPANTS

Youth with early maternal deprivation and comparison youth.

DEVELOPMENTAL MEASUREMENT

Measured amygdala–medial-prefrontal functional coupling, compared across development.

STIMULUS / EXPOSURE

Emotional-processing fMRI paradigm.

OUTCOME ACTUALLY MEASURED

Functional connectivity and anxiety associations.

INTERACTION OR STRATIFICATION

Early environment was associated with a developmentally atypical circuit pattern.

Finding: Previously institutionalized children showed earlier emergence of an adolescent-like coupling pattern. Some relative anxiety reduction coexisted with higher anxiety in the exposed group.

Sex / gender interpretation: No sex-specific response rule is taken from this study.

Limits: A circuit-specific observation, not uniformly faster or healthier brain maturation. Does not permit inferring circuit maturity from a person’s history.

02 INPUTS, NOT DIAGNOSES

Your observation ledger

Every current response is shown, including the unknowns. A skipped or unreviewed question is not a negative answer.

Observation	Type	Your response
Signal scout	strength	Often
Idea weaver	strength	Often
Deep diver	strength	Often
Adaptive navigator	strength	Not reviewed
Connection builder	strength	Not reviewed
System builder	strength	Sometimes
Too much input	friction	Sometimes
Thought loops	friction	Not reviewed
Switching cost	friction	Often
Unclear expectations	friction	Not reviewed
Social energy drain	friction	Not reviewed
Recovery takes time	friction	Not reviewed
Trusted support	Current resource	Often
Restorative routine	Current resource	Sometimes
A workable environment	Current resource	Often
Choice and boundaries	Current resource	Not reviewed

7 present-day observations remain unknown (not reviewed, not sure, or skipped). Rarely / not now is a time-specific response, not proof that a capacity is absent.

What helped earlier

- Dependable adult
- Trusted friendship
- Feeling capable

What helps now

- Trusted support · Often
- Restorative routine · Sometimes
- A workable environment · Often

Childhood and current supports are not treated as interchangeable, and no numerical protective effect is estimated.

Privacy: Adversity and prenatal responses, including developmental context and calendar ages, are excluded by default. Reported childhood supports are included. Enabling the private appendix is a separate choice. Present-day observations and past/current support information are still sensitive. Share thoughtfully.

03 CONTEXTUAL INTERPRETATION

Possible strengths and friction



Often · self-reported

Signal scout

Noticing changes and checking patterns.

A possible asset in observation, quality checks, or spotting a change early.

Try noticing: Notice one detail, then check whether your interpretation was accurate.



Often · self-reported

Idea weaver

Generating and connecting possibilities.

A possible asset when a problem benefits from several perspectives.

Try noticing: Write down three approaches before choosing one small experiment.



Often · self-reported

Deep diver

Sustained interest-led attention.

A possible asset for learning or detailed work in a supportive setting.

Try noticing: Choose one bounded focus session with a visible stopping point.



Sometimes · self-reported

System builder

Structuring and sequencing a task.

A possible asset for coordinating work or making a process clearer.

Try noticing: Define the next visible action and what done looks like.



Sometimes · self-reported

Too much input

Competing input can make a situation demanding.

Try noticing: Experiment with reducing one input or handling one demand at a time.



Often · self-reported

Switching cost

Transitions between tasks can take effort.

Try noticing: Try an advance transition cue and leave a note about where to resume.

04 COMBINATIONS, MODERATION & UNCERTAINTY

When more than one pattern is present

Co-occurrence means two things are reported together. Synergy requires evidence that their combined effect differs from the appropriate comparison or additive expectation on a specified outcome and scale. Co-occurrence alone does not demonstrate synergy.

The following combinations come only from current strength observations. Their possible benefits and costs are editorial hypotheses. No causal or anatomical claim follows.

UNTESTED COMBINATION HYPOTHESIS

Pattern laboratory

Signal scout (Often) + Idea weaver (Often)

Careful noticing plus idea generation may help you form useful alternative explanations.

Possible kryptonite: A plausible trade-off is making too many connections, including false alarms.

Context: Most useful when there is time to check a pattern against evidence.

Reality check: Record a prediction before the outcome. Count the misses as well as the hits.

Reported friction: Too much input

Reported support: Trusted support, A workable environment

No study in this registry validates this named combination or its neural mechanism. Neither component nor their combination establishes any anatomical change.

UNTESTED COMBINATION HYPOTHESIS

Focused builder

Deep diver (Often) + System builder (Sometimes)

Interest-led concentration plus clear steps may help turn a complex project into finished work.

Possible kryptonite: A plausible trade-off is investing deeply in one path and finding transitions difficult.

Context: May work best with a bounded goal and an explicit stopping point.

Reality check: Compare one planned session with an unplanned session on a similar task; note other differences.

Reported friction: Switching cost

Reported support: Restorative routine

No study in this registry validates this named combination or its neural mechanism. Neither component nor their combination establishes any anatomical change.



04B WHAT COMBINATION RESEARCH CAN SUPPORT

Interactions and protective factors

Direct combined-exposure research is a different category

The registry includes Puetz et al., comparing abuse-only, neglect-only, combined and comparison groups on a threat-processing task. Its group findings were not a simple sum of separate patterns. That result does not validate a “meta-power,” prove a brain deformation, or let this app predict an individual response.

See the complete Puetz study card in the primary-study library below. It is shown for every reader as methods context, not because we have assigned you to an exposure group.

Protective factors: possible modifiers, not a subtraction

Caregiving, relationships and current resources are separate observations. Whether a particular factor modifies a particular association depends on the population, timing, design and outcome. The Luby caregiving study and Bick foster-care trial below offer different kinds of evidence; neither supplies a universal buffering percentage for these answers.

Unknown here: whether a combination changes your performance, whether supports modify that effect, its direction or magnitude, and any underlying neural mechanism.



05 A SHARED EDUCATIONAL MODEL

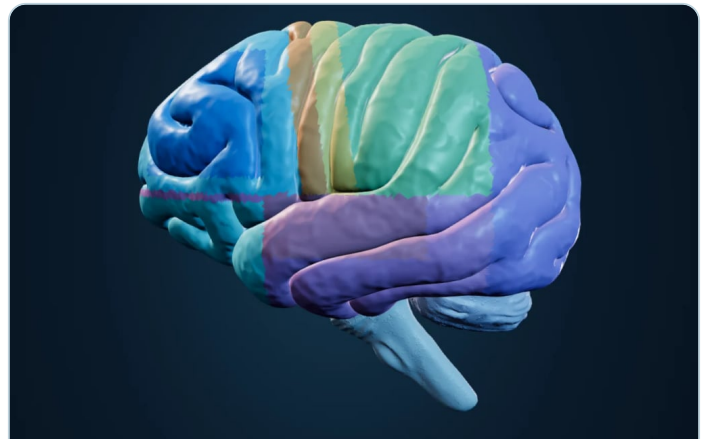
The brain, separated into readable systems

Locations and functions are broad teaching summaries. A region participates in multiple networks; it is not a standalone “power center.”



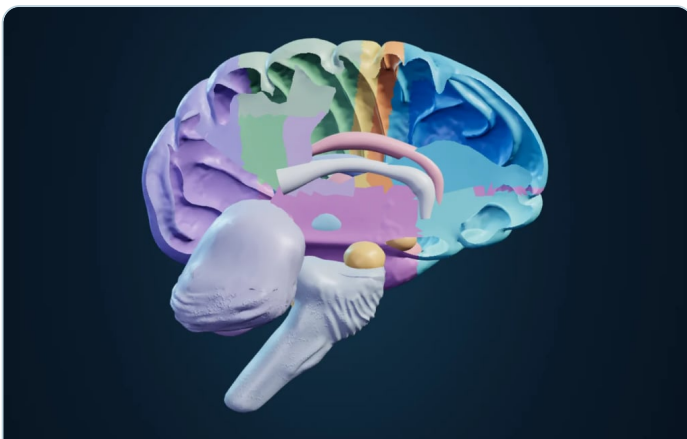
Left lateral cortical surface

Generic model · not a scan



Right lateral cortical surface

Generic model · not a scan



Medial cutaway

Generic model · not a scan



Separated hemispheres and internal guides

Generic model · not a scan

The model has 26 selectable teaching topics and 47 meshes. Cortical partitions have geometric, not atlas-registered boundaries. Internal structures are schematic. These are not personal scans, measured changes, injuries, or segmented research clusters.

Artwork adapted in Blender from “Brain - with labeled parts” by AbdulMuhaymin, CC BY 4.0. Original: sketchfab.com/3d-models/brain-with-labeled-parts-28c8971e11334e8b97a2a0d6235992e8. Geometric teaching partitions and schematic guides were added. Original authors are not represented as endorsing this tool.

● Frontal lobe

Where: Front of each cerebral hemisphere.

Role: Supports planning and coordinated behavior with other brain networks.

Studied associations: Hair et al. (2015) (Regional gray-matter volume and academic assessments)

Geometric teaching boundary on a generic surface, not a registered anatomical atlas. Follow the blue anterior surface. Named frontal subdivisions have their own colors.

Anatomy source: BrainFacts / Society for Neuroscience — Major brain landmarks

● Dorsolateral prefrontal cortex

Where: Upper, outer prefrontal cortex.

Role: A contributor to executive-control networks for goal-directed tasks.

Studied associations: Puetz et al. (2020; online 2019) (Neural response to threatening facial cues)

Geometric teaching boundary on a generic surface, not a registered anatomical atlas. This patch marks an approximate teaching location, not a measured functional boundary.

Anatomy source: Seeley et al. (2007) — Salience and executive-control networks

● Orbitofrontal cortex

Where: Underside of the frontal lobe, above the eye sockets.

Role: Helps represent expected outcomes and their changing value.

Studied associations: Wang et al. (2026; online March 5) (Age- and sex-stratified deviations in volume, thickness and surface area)

Geometric teaching boundary on a generic surface, not a registered anatomical atlas. Rotate toward the underside or isolate this region to see its position.

Anatomy source: Gottfried et al. (2003) — Predictive reward value

● Primary motor cortex

Where: Precentral gyrus, in front of the central sulcus.

Role: Contributes commands for voluntary movement.

Studied associations: No specific adversity study is assigned in this curated registry. This is not evidence that no research exists.

Geometric teaching boundary on a generic surface, not a registered anatomical atlas. Compare the orange strip with the sensory strip just behind it. Strips are illustrative.

Anatomy source: Purves et al. — Primary motor cortex

● Primary somatosensory cortex

Where: Postcentral gyrus, behind the central sulcus.

Role: Processes body sensation, including touch and position.

Studied associations: Heim et al. (2013) (Cortical thickness)

Geometric teaching boundary on a generic surface, not a registered anatomical atlas. The yellow strip is posterior to the motor strip; a full body map is not modeled.

Anatomy source: Raju & Tadi — Somatosensory cortex

● Parietal lobe

Where: Upper posterior cerebrum, behind the frontal lobe.

Role: Integrates sensory information and spatial relationships.

Studied associations: Puetz et al. (2020; online 2019) (Neural response to threatening facial cues); Wang et al. (2026; online March 5) (Age- and sex-stratified deviations in volume, thickness and surface area); Lebel et al. (2012) (Cortical volume trajectories across repeated scans)

Geometric teaching boundary on a generic surface, not a registered anatomical atlas. Find the green upper surface behind the sensory strip.

Anatomy source: BrainFacts / Society for Neuroscience — Major brain landmarks

● Temporal lobe

Where: Lateral lower cerebrum, below the lateral fissure.

Role: Supports hearing, memory and aspects of language processing.

Studied associations: Hair et al. (2015) (Regional gray-matter volume and academic assessments)

Geometric teaching boundary on a generic surface, not a registered anatomical atlas. The purple side region is a lobe, not one single language or social center.

[Anatomy source: BrainFacts / Society for Neuroscience — Major brain landmarks](#)

● Occipital lobe

Where: Back of each cerebral hemisphere.

Role: Processes visual information.

Studied associations: Wang et al. (2026; online March 5) (Age- and sex-stratified deviations in volume, thickness and surface area)

Geometric teaching boundary on a generic surface, not a registered anatomical atlas. Rotate to the rear to distinguish the occipital lobe from the cerebellum below.

[Anatomy source: BrainFacts / Society for Neuroscience — Major brain landmarks](#)

● Auditory association cortex

Where: Upper temporal surface near the lateral fissure.

Role: Interpreting sounds and language-related auditory information.

Studied associations: Tomoda et al. (2011; online 2010) (Regional gray-matter volume)

A broad superior-temporal teaching patch; not the exact cluster from a study.

[Anatomy source: Johns Hopkins Medicine — Brain anatomy](#)

● Visual cortex (medial occipital)

Where: Back and inner surface of the occipital lobe.

Role: Visual processing within multiple interacting visual areas.

Studied associations: Tomoda et al. (2012) (Gray-matter volume and cortical thickness)

A broad medial-occipital teaching patch, not a precise V2 or lingual-gyrus segmentation.

[Anatomy source: Johns Hopkins Medicine — Brain anatomy](#)

● Posterior cingulate cortex

Where: Inner posterior cortex above the corpus callosum.

Role: A contributor to the default-mode network; its roles depend on task and context.

Studied associations: Thorsen et al. (2025, December 18) (Cortical thickness and T1w/T2w signal ratio)

A small medial teaching patch, not a precise posterior-cingulate or isthmus atlas parcel.

[Anatomy source: Greicius et al. \(2003\) — Default-mode functional connectivity](#)

● Precuneus

Where: Medial parietal surface above the posterior cingulate area.

Role: Participates in higher-order visual and awareness-related processing.

Studied associations: No specific adversity study is assigned in this curated registry. This is not evidence that no research exists.

An approximate medial teaching area.

[Anatomy source: Kjaer et al. \(2001\) — Precuneus-prefrontal activity](#)

● Amygdala

Where: Medial temporal region, near the front of the hippocampus.

Role: Salience, learning and responses to significant events.

Studied associations: McCrory et al. (2011) (Response to angry versus other facial expressions); Puetz et al. (2020; online 2019) (Neural response to threatening facial cues); Chen, Lopez-Quintero & Elton (2026; online 2025) (Connections between subcortical regions and distributed networks); Rifkin-Graboi et al. (2013) (Amygdala diffusion measures and volume)

Not simply a fear switch; shown as paired schematic structures.

[Anatomy source: Johns Hopkins Medicine — Brain anatomy](#)

● Hippocampus

Where: Medial temporal region in each hemisphere.

Role: Learning, memory and contextual relationships.

Studied associations: Hair et al. (2015) (Regional gray-matter volume and academic assessments); Luby et al. (2012) (Observed maternal support and later hippocampal volume); Chen, Lopez-Quintero & Elton (2026; online 2025) (Connections between subcortical regions and distributed networks); Wang et al. (2026; online March 5) (Age- and sex-stratified deviations in volume, thickness and surface area)

Curved teaching shape; hippocampal subfields are not individually segmented.

[Anatomy source: Johns Hopkins Medicine — Brain anatomy](#)

● Anterior cingulate cortex

Where: Medial frontal surface arching above the front of the corpus callosum.

Role: Monitoring, regulation and coordination within networks.

Studied associations: Thorsen et al. (2025, December 18) (Cortical thickness and T1w/T2w signal ratio)

The cingulate is cortex, not a deep nucleus. This arch is a schematic medial-cortex guide.

[Anatomy source: Seeley et al. \(2007\) — Saliency and executive-control networks](#)

● Insula

Where: Cortex buried inside the lateral fissure.

Role: Integrates bodily and other salient information.

Studied associations: McCrory et al. (2011) (Response to angry versus other facial expressions)

The insula is folded cortex, shown schematically beneath the outer surface.

[Anatomy source: Menon & Uddin \(2010\) — A network model of insula function](#)

● Thalamus

Where: Paired structures near the center of the brain.

Role: Routing and coordinating sensory and other information.

Studied associations: No specific adversity study is assigned in this curated registry. This is not evidence that no research exists.

Many thalamic nuclei are simplified into one teaching form per side.

[Anatomy source: Johns Hopkins Medicine — Brain anatomy](#)

● Hypothalamus

Where: Below the thalamus near the brain base.

Role: Homeostatic and hormonal regulation.

Studied associations: No specific adversity study is assigned in this curated registry. This is not evidence that no research exists.

A small midline teaching form; no hormone concentration is inferred.

[Anatomy source: Johns Hopkins Medicine — Brain anatomy](#)

● Periaqueductal gray

Where: Around the cerebral aqueduct in the midbrain.

Role: Contributes to pain modulation and defensive responses.

Studied associations: No specific adversity study is assigned in this curated registry. This is not evidence that no research exists.

Highly simplified ring-like guide; not the entire midbrain.

[Anatomy source: Bandler & Shipley \(1994\) — PAG and defensive behavior](#)

● Corpus callosum

Where: Midline white-matter arch between the cerebral hemispheres.

Role: Communication between hemispheres.

Studied associations: Bick et al. (2015) (White-matter diffusion measures)

Two joined schematic halves illustrate a continuous commissure, not measured fiber pathways.

[Anatomy source: Johns Hopkins Medicine — Brain anatomy](#)

● Cerebellar vermis

Where: Midline portion of the cerebellum.

Role: Coordination and regulation within cerebellar networks.

Studied associations: No specific adversity study is assigned in this curated registry. This is not evidence that no research exists.

Schematic vermis shown separately from cerebellar hemispheres.

[Anatomy source: Johns Hopkins Medicine — Brain anatomy](#)

● Ventral striatum / nucleus accumbens guide

Where: Ventral forebrain below the caudate head.

Role: Reward-related learning and motivated behavior.

Studied associations: Hanson, Hariri & Williamson (2015) (Reward-related ventral striatal activity); Chen, Lopez-Quintero & Elton (2026; online 2025) (Connections between subcortical regions and distributed networks)

The small accumbens-centered guide is not the full ventral striatum or a pleasure meter.

[Anatomy source: Purves et al. — Projections to the basal ganglia](#)

● Caudate nucleus

Where: Curved structure around the lateral ventricle, medial to the putamen.

Role: Contributes to action selection and learning.

Studied associations: Burke Quinlan et al. (2018 online; 2020 issue) (Subcortical volume change)

A schematic curved nucleus; no neuronal counts or personal volume are represented.

[Anatomy source: Purves et al. — Projections to the basal ganglia](#)

● Putamen

Where: Lateral striatum, outside the caudate and thalamus.

Role: Action selection and motor-learning networks.

Studied associations: Burke Quinlan et al. (2018 online; 2020 issue) (Subcortical volume change); Wang et al. (2026; online March 5) (Age- and sex-stratified deviations in volume, thickness and surface area)

A schematic bilateral shape, distinct from the ventral-striatum guide.

[Anatomy source: Purves et al. — Projections to the basal ganglia](#)

● Cerebellar hemispheres

Where: Below the posterior cerebrum.

Role: Coordination, timing and learning in distributed networks.

Studied associations: No specific adversity study is assigned in this curated registry. This is not evidence that no research exists.

Original licensed surface retained; no precise cerebellar lobular atlas.

[Anatomy source: Johns Hopkins Medicine — Brain anatomy](#)

● Brainstem

Where: Below the cerebrum, continuous with the spinal cord.

Role: Contains pathways and centers supporting essential body functions.

Studied associations: No specific adversity study is assigned in this curated registry. This is not evidence that no research exists.

General teaching anatomy; not a localization of a respondent's symptoms.

[Anatomy source: Johns Hopkins Medicine — Brain anatomy](#)

06 FULL CURATED PRIMARY-STUDY REGISTRY

What the studies actually found

All 15 registry entries are included so the library does not reveal which history answers you selected. Each entry keeps the measurement, population, finding and limitation together. This is not an exhaustive literature review.

Cross-sectional structural MRI

Tomoda et al. (2011; online 2010)

DOI: 10.1016/j.neuroimage.2010.05.027

PARTICIPANTS

40 young adults aged 18–25: 21 parental-verbal-aggression participants, 19 controls.

WHAT WAS MEASURED

Regional gray-matter volume

Finding: The exposed group had greater volume in the left superior temporal gyrus, an auditory association region.

Auditory association · Left

Do not infer: Small selected sample; parental aggression differs from mistreatment by any adult. Greater volume does not mean better function, and this cannot estimate an individual brain.

Age and timing: Childhood exposure recalled in young adulthood; not a universal age cutoff.

Source reviewed: Primary full text.

Cross-sectional structural MRI

Tomoda et al. (2012)

DOI: 10.1371/journal.pone.0052528

PARTICIPANTS

52 adults aged 18–25: 22 who visually witnessed domestic violence and 30 controls.

WHAT WAS MEASURED

Gray-matter volume and cortical thickness

Finding: Lower right lingual-gyrus volume and lower thickness in visual areas including bilateral V2 were reported.

Visual cortex · Bilateral guide

Do not infer: Selected sample, retrospective exposure and overlapping adversity limit generalization. This does not show that every witness has visual-cortex changes.

Age and timing: The study's exploratory timing analysis emphasized 11–13 years; this is not a multiplier, a safe/unsafe boundary, or a validated threshold.

Source reviewed: Primary full text.

Cross-sectional structural MRI

Heim et al. (2013)

DOI: 10.1176/appi.ajp.2013.12070950

PARTICIPANTS

51 medically healthy adult women.

WHAT WAS MEASURED

Cortical thickness

Finding: Sexual-abuse scores were associated with thinner cortex in a left somatosensory genital-representation field; emotional abuse had different regional associations.

Sensory strip · Left

Do not infer: This small study cannot establish causation, generalize to every sex or age, or predict sensation or sexual function. The model shows a broad somatosensory teaching strip, not that precise field.

Age and timing: Retrospective childhood report; no exposure-age threshold is applied.

Source reviewed: Primary full text.

Longitudinal task fMRI

Hanson, Hariri & Williamson (2015)

DOI: 10.1016/j.biopsych.2015.05.010

PARTICIPANTS

106 participants initially aged 11–15, rescanned about two years later.

WHAT WAS MEASURED

Reward-related ventral striatal activity

Finding: Greater emotional neglect was associated with a smaller developmental increase in reward-related ventral-striatum response.

Ventral striatum · Bilateral guide

Do not infer: Task activity is not tissue loss or everyday motivation. Observational mediation does not establish a causal chain or a person's depression risk.

Age and timing: 11–15 describes ages at the first scan, not a proven sensitive period for neglect.

Source reviewed: Primary abstract.

Longitudinal structural MRI

Burke Quinlan et al. (2018 online; 2020 issue)

DOI: 10.1038/s41380-018-0297-9

PARTICIPANTS

682 adolescents in IMAGEN; MRI at approximately 14 and 19 years.

WHAT WAS MEASURED

Subcortical volume change

Finding: Victimization trajectories interacted with left putamen volume change; indirect associations with anxiety were reported for left putamen and caudate.

Putamen · Left | Caudate · Left

Do not infer: Peer victimization was studied as a broader pattern, not isolated physical versus emotional bullying. Volume is not a measure of personal worth, resilience or symptoms.

Age and timing: Scan ages and exposure trajectories are distinct; no threshold is imported into this questionnaire.

Source reviewed: Primary abstract.

Task fMRI group comparison

McCrory et al. (2011)

DOI: 10.1016/j.cub.2011.10.015

PARTICIPANTS

Children with documented family-violence exposure compared with non-exposed peers.

WHAT WAS MEASURED

Response to angry versus other facial expressions

Finding: The exposed group showed greater anterior-insula and amygdala responses to angry faces.

Amygdala · Bilateral guide | Insula · Bilateral guide

Do not infer: The exposure group included family violence rather than this one question in isolation. Reactivity during a task is not a brain-injury scan or a prediction of permanent hypervigilance.

Age and timing: Childhood group comparison; not a prenatal or infancy result.

Source reviewed: Primary abstract / publisher summary.

Propensity-matched task fMRI comparison

Puetz et al. (2020; online 2019)

DOI: 10.1017/S003329171900134X

PARTICIPANTS

414 participants across abuse-only, neglect-only, combined and comparison groups.

WHAT WAS MEASURED

Neural response to threatening facial cues

Finding: Abuse was linked to ventral-amygdala reactivity; neglect involved dorsal amygdala and frontoparietal networks. Combined exposure showed lower responses in several regions.

Amygdala · Bilateral guide | dlPFC · Bilateral guide | Parietal lobe · Bilateral guide

Do not infer: The combined pattern was not the sum of the separate patterns. These are broad exposure groups, not isolated effects of individual survey items; no activation estimate follows from an answer.

Age and timing: No age-band weights or brain-response thresholds are imported.

Source reviewed: Primary abstract; author-name corrigendum checked. [Published correction](#).

Randomized foster-care intervention with diffusion MRI

Bick et al. (2015)

DOI: 10.1001/jamapediatrics.2014.3212

PARTICIPANTS

Diffusion-MRI analysis: 69 children aged 8–11—23 foster-care, 26 institutional-care and 20 never-institutionalized participants.

WHAT WAS MEASURED

White-matter diffusion measures

Finding: Diffusion differences involved the corpus-callosum body and other tracts. The foster-care group still differed from the never-institutionalized group on callosal measures.

Corpus callosum · Bilateral guide

Do not infer: Severe institutional deprivation is not equivalent to any single instance of unmet needs or supervision. Randomization concerned foster-care placement, not assignment to maltreatment.

Age and timing: Infancy exposure and school-age imaging are different time points. No personal tract damage is inferred.

Source reviewed: Primary abstract / full publisher text.

Longitudinal structural MRI cohort

Hair et al. (2015)

DOI: 10.1001/jamapediatrics.2015.1475

PARTICIPANTS

389 children and young adults aged 4–22; 823 scans.

WHAT WAS MEASURED

Regional gray-matter volume and academic assessments

Finding: Household income relative to family needs was associated with development of frontal, temporal and hippocampal regions.

Frontal lobe · Bilateral guide Temporal lobe · Bilateral guide Hippocampus · Bilateral guide

Do not infer: Household income was the exposure—not this unmet-needs question. Poverty is not a diagnosis, proof of neglect, or a fixed limit on ability. Multiple environmental influences may contribute.

Age and timing: Participant scan ages do not define sensitive periods for a respondent.

Source reviewed: Primary abstract.

Prospective caregiving study with later structural MRI

Luby et al. (2012)

DOI: 10.1073/pnas.1118003109

PARTICIPANTS

Preschool cohort including depressed and nondepressed children, imaged at school age.

WHAT WAS MEASURED

Observed maternal support and later hippocampal volume

Finding: More observed early maternal support predicted larger later hippocampal volume, with differences by preschool depression status.

Hippocampus · Bilateral guide

Do not infer: Observed maternal behavior is not equivalent to a retrospective Yes about any dependable adult. Larger is not universally better; this does not quantify buffering or growth in you.

Age and timing: Early observations preceded school-age scans. Findings are support-context reading, never subtracted from adversity.

Source reviewed: Primary abstract.

Longitudinal resting-state fMRI / ABCD

Chen, Lopez-Quintero & Elton (2026; online 2025)

DOI: 10.1016/j.jaac.2025.04.005

PARTICIPANTS

3,321 racial/ethnic minority youths; two-year connectivity changes.

WHAT WAS MEASURED

Connections between subcortical regions and distributed networks

Finding: Perceived racial discrimination was associated with changes involving nucleus accumbens, amygdala and hippocampal connectivity.

■ Ventral striatum · Bilateral guide ■ Amygdala · Bilateral guide ■ Hippocampus · Bilateral guide

Do not infer: Our broad identity-discrimination item does not establish racial discrimination. Connectivity is not volume or damage. This analysis is observational and does not imply biological racial categories.

Age and timing: Two-year developmental change, not a validated age-triggered effect in this questionnaire.

Source reviewed: Primary abstract.

Multisite structural MRI mega-analysis

Wang et al. (2026; online March 5)

DOI: 10.1016/j.biopsych.2026.02.016

PARTICIPANTS

3,711 participants across 25 ENIGMA MDD/PTSD sites and comparison groups.

WHAT WAS MEASURED

Age- and sex-stratified deviations in volume, thickness and surface area

Finding: Associations differed across regions, ages and sexes. Some structures were larger, others smaller; no significant effects were detected in the pediatric group.

■ Hippocampus · Bilateral guide ■ Putamen · Bilateral guide ■ Orbitofrontal · Bilateral guide ■ Occipital lobe · Bilateral guide
■ Parietal lobe · Bilateral guide

Do not infer: The pediatric null findings had power limitations and do not mean childhood adversity has no effects. Broad abuse/neglect measures and cross-sectional MRI cannot reconstruct individual changes or isolate our questions.

Age and timing: Age at scan is not age of exposure. No sex-specific or age-specific personal score is generated.

Source reviewed: Primary publisher full text.

Longitudinal structural MRI / ABCD

Thorsen et al. (2025, December 18)

DOI: 10.1017/S0033291725102651

PARTICIPANTS

ABCD children aged about 9–13; 1,923 followed with paired scans.

WHAT WAS MEASURED

Cortical thickness and T1w/T2w signal ratio

Finding: Parental acceptance related to posterior-cingulate thickness at baseline and greater isthmus-cingulate thinning over time. Household abuse related to smaller T1w/T2w changes, including anterior cingulate.

Posterior cingulate · Bilateral guide | Anterior cingulate · Bilateral guide

Do not infer: Thinner cortex is not automatically injury. T1w/T2w is nonspecific; the posterior-cingulate model is not an exact isthmus parcel. ABCD analyses may share participants rather than independently replicate one another.

Age and timing: Study ages describe observed developmental trajectories, not a universal sensitive-period rule.

Source reviewed: Primary publisher full text.

Prospective prenatal assessment / neonatal MRI

Rifkin-Graboi et al. (2013)

DOI: 10.1016/j.biopsych.2013.06.019

PARTICIPANTS

157 newborns imaged at 6–14 days; maternal symptoms assessed at 26 weeks of pregnancy.

WHAT WAS MEASURED

Amygdala diffusion measures and volume

Finding: Higher prenatal depression scores were associated with lower fractional anisotropy and axial diffusivity in the right amygdala, but not a volume difference.

Amygdala · Right

Do not infer: A retrospective family account is not the study's measured prenatal symptoms. Diffusion is not tissue damage. The adult-shaped teaching brain is not a model of neonatal anatomy or a prediction for the respondent.

Age and timing: Pregnancy measurement and newborn scan timing are recorded separately; no post-birth age band or personal risk score is inferred.

Source reviewed: Primary abstract.

Longitudinal structural MRI

Lebel et al. (2012)

DOI: 10.1523/JNEUROSCI.1161-12.2012

PARTICIPANTS

Children and youth with heavy prenatal alcohol exposure and unexposed comparison participants.

WHAT WAS MEASURED

Cortical volume trajectories across repeated scans

Finding: Developmental trajectories differed especially in posterior regions, including parietal cortex, in the heavy-exposure group.

| Parietal lobe · Bilateral guide

Do not infer: The app's Yes/No pregnancy item does not measure dose or establish heavy exposure. These data cannot diagnose fetal alcohol spectrum disorder or estimate an individual's brain trajectory.

Age and timing: Exposure during pregnancy and the later scan ages are distinct. This stays in prenatal context, never in the childhood adversity count.

Source reviewed: Primary abstract.

07 COVERAGE & ASSUMPTIONS

Question-to-research audit

These are all questionnaire domains—not your answers. Domain and related describe exposure fit, not a clinical certainty grade. Missing mappings remain visible.

Physical harm related

Family-violence and broader abuse samples are relevant, but do not isolate this one question.

Linked primary studies: McCrory et al. (2011); Puetz et al. (2020; online 2019); Wang et al. (2026; online March 5)

Threats of harm related

Threat-only experience was not isolated in this family-violence study.

Linked primary studies: McCrory et al. (2011)

Verbal mistreatment domain

This study examined parental verbal aggression; the question also permits other adults. The left auditory-association location is more specific than “temporal lobe.”

Linked primary studies: Tomoda et al. (2011; online 2010)

Emotional support domain

Emotional neglect has specific reward-response research. Broader neglect findings and mixed exposures are shown separately.

Linked primary studies: Hanson, Hariri & Williamson (2015); Puetz et al. (2020; online 2019); Wang et al. (2026; online March 5)

Basic material needs related

Income-based poverty research is not a direct measure of food, clothing or the cause of unmet needs.

Linked primary studies: Hair et al. (2015)

Access to needed care unmapped

No reviewed study here isolates repeated failure to obtain care from illness, financial access and other circumstances. A specific region is not assigned.

No region assignment in the current review.

Safe supervision related

Institutional deprivation is much broader than this supervision item. Read it as related context, not a one-to-one mapping.

Linked primary studies: Bick et al. (2015)

Household alcohol problems unmapped

The household-alcohol domain is not the same as prenatal alcohol exposure, the respondent’s own alcohol use or inherited familial risk. These cannot be substituted to assign a region.

No region assignment in the current review.

Household drug use unmapped

Drug type, prenatal exposure, caregiving and the respondent’s own use are different exposures. This item alone does not specify a brain-study match.

No region assignment in the current review.

Household incarceration unmapped

The reviewed incarceration sources describe health and behavioral outcomes, not an isolated regional brain effect. An important experience can remain unmapped.

No region assignment in the current review.

Household mental health unmapped

A household member’s mental health, the care environment and familial risk are distinct. A diagnosis in someone else does not assign a brain effect to the respondent.

No region assignment in the current review.

Suicide attempt in the household unmapped

A household suicide attempt has its own meaning and context. The selected sources do not justify an isolated region assignment from this Yes/No item.

No region assignment in the current review.

Caregiver separation unmapped

Separation or divorce is not equivalent to severe institutional deprivation, bereavement or abuse; those imaging findings are not transferred to this item.

No region assignment in the current review.

Witnessed caregiver violence domain

The visual study required visually witnessed interparental violence. The separate family-violence fMRI study measures threat response, not cortical thickness.

Linked primary studies: Tomoda et al. (2012); McCrory et al. (2011)

Peer emotional mistreatment domain

The source measured broader repeated peer victimization, not a uniquely emotional-bullying neural signature.

Linked primary studies: Burke Quinlan et al. (2018 online; 2020 issue)

Peer physical bullying domain

The source measured broader peer victimization. It does not establish a separate physical-bullying effect or an additional effect to add to emotional bullying.

Linked primary studies: Burke Quinlan et al. (2018 online; 2020 issue)

Sexual boundaries domain

The specific sensory finding came from adult women and a much more detailed abuse measure; other samples may differ.

Linked primary studies: Heim et al. (2013); Puetz et al. (2020; online 2019)

Identity-based mistreatment related

The study concerns racial discrimination. This broader identity-discrimination question must not be treated as proof of that narrower exposure.

Linked primary studies: Chen, Lopez-Quintero & Elton (2026; online 2025)

Medical experiences unmapped

Serious illness, painful treatment and perceived medical threat differ biologically. No particular illness or neurological injury is assumed from this question.

No region assignment in the current review.

Caregiver death unmapped

Bereavement, prolonged grief, caregiver disruption and depression are not interchangeable. No reviewed study here supports assigning a regional change to every bereaved child.

No region assignment in the current review.

Disaster exposure unmapped

Disaster studies often involve particular events and selected symptomatic samples. A generic exposure answer is not enough to infer PTSD-related brain findings.

No region assignment in the current review.

Before birth: reported depression related

Measured prenatal depression and neonatal diffusion findings are shown as separate research context. The reported history does not activate a personal brain prediction.

Linked primary studies: Rifkin-Graboi et al. (2013)

Before birth: reported alcohol exposure related

The source studied heavy prenatal exposure. A Yes without dose or verified details cannot inherit that group's brain-development findings; this is context only.

Linked primary studies: Lebel et al. (2012)

Dependable adult related

Observed maternal support and parental acceptance are related to, but narrower than, any dependable adult. These support studies do not cancel adversity or estimate growth.

Linked primary studies: Luby et al. (2012); Thorsen et al. (2025, December 18)

Trusted friendship unmapped

Friendship is an endorsed support, not a biomarker of a particular brain region. The population support study does not provide an individual neural estimate.

No region assignment in the current review.

Belonging unmapped

Different sources of belonging should not be collapsed into a single regional neural effect. Keep this as a self-reported support.

No region assignment in the current review.

Feeling capable unmapped

A skill or feeling of competence does not identify a neural superpower or larger brain structure. This remains an explicit personal-strength reflection.

No region assignment in the current review.

Grounding routines unmapped

Predictable routines may feel helpful; this item is not a sleep, stress-hormone or hypothalamus assessment.

No region assignment in the current review.

Questionnaire and context sources

CDC BRFSS (2023), Module 24, pp. 105–108 · <https://www.cdc.gov/brfss/questionnaires/pdf-ques/2023-BRFSS-Questionnaire-508.pdf#page=105>

Teicher & Parigger (2015) · <https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0117423>

Teicher & Samson (2016) · <https://pubmed.ncbi.nlm.nih.gov/26831814/>

Anda et al. (2002) · <https://pubmed.ncbi.nlm.nih.gov/12161676/>

Turney (2014) · <https://pubmed.ncbi.nlm.nih.gov/25138199/>

Wildeman & Turney (2014) · <https://pubmed.ncbi.nlm.nih.gov/24719275/>

Bethell et al. (2019) · <https://pubmed.ncbi.nlm.nih.gov/31498386/>

Interpretation checklist

Do not convert task activity into tissue volume, connectivity into damage, a retrospective age range into a sensitive-period multiplier, or a supportive relationship into a percentage reduction. Study populations differ; shared cohorts do not automatically provide independent replication. Null and conflicting findings are retained. Current self-report is not proof of objectively superior ability.

Versions: cc-reflection-5.0 · cc-region-evidence-5.0 · cc-insights-1.0 · cc-reports-7.0 · cc-development-1.0 · cc-context-1.0.
Rule order is deterministic. No generative AI or cloud scoring is used to produce this report.